

Kaustubh Sridhar

Research Scientist @ Google Deepmind

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Education

- Aug 2019 - May 2025 **University of Pennsylvania,** *Philadelphia, PA.*
PhD Candidate, Electrical and Systems Engineering, GPA: 3.94/4.
ASSET and **PRECISE** Center
Thesis Title: Training Adaptive and Sample-Efficient Autonomous Agents
Thesis Committee: [Prof Dinesh Jayaraman](#), [Prof Insup Lee](#), [Prof George Pappas](#), [Prof Nikolai Matni](#), [Prof Dorsa Sadigh](#)
- Jul 2015 - May 2019 **Indian Institute of Technology Bombay,** *Mumbai, India.*
Bachelor Of Technology (with Honors) In Aerospace Engineering, GPA: 9.07/10.
Minor in Systems and Control Engineering Class Rank 2.

Research Interests

I am interested in creating adaptive generalist agents that are parameter- and sample-efficient, for the digital and physical worlds. Towards this goal, I have worked on generative models, in-context learning, deep reinforcement and imitation learning, and robust deep learning. My recent work on the [SIMA agent that can play any 3D virtual world, adding in-context adaptability to pre-trained VLAs \(RICL\)](#), and a [retrieval-augmented generalist agent \(REGENT\)](#) directly aims for this goal. For the physical world in particular, I believe that world models are on the critical path to reach embodied AGI. For this reason, I have worked on bringing [Genie 3](#) to robotics.

Work Experience

- May 2025 - Present **Google Deepmind, Research Scientist,** *Toronto, Canada*
○ **SIMA Project** [[2025C](#), [2025B](#), [videos](#)]
- Post-trained Gemini flash-sized models to predict keyboard and mouse actions to complete any long-horizon task in virtual worlds.
- Spearheaded SIMA's autonomous self-improvement in unseen worlds, to human-level performance, with a Gemini-based reward function and a Gemini-based task-setter.
○ **Genie Robotics Project:**
- Post-trained [Genie 3](#) on robotics datasets to create a world model for humanoid and tabletop (aloha) robots.
- Genie Robotics enabled planning with Gemini Robotics VLAs, improving their success rate (almost 2x) with more test-time compute on long-horizon and long-tail tasks.
- Aug 2019 - May 2025 **University of Pennsylvania, PhD Candidate,** *Philadelphia, PA.*
Advised by [Prof Insup Lee](#).
Closely collaborated with [Prof Dinesh Jayaraman](#), [Prof Oleg Sokolsky](#).
○ Post-trained a VLA to add in-context adaptability to boost its performance on new robotic tasks with unseen objects, motions, and scenes [[2025A](#), [videos](#)].
○ Pretrained a generalist agent that can generalize to unseen robotics and game-playing environments via retrieval-augmentation & in-context learning [[2024A](#), [videos](#)].
○ Strengthened imitation learning with any neural network – diffusion models, transformers, or MLPs, via a novel semi-parametric model class called the MCNN [[2023B](#), [videos](#)].
○ Created a tool for guaranteed conformance of generative models to constraints [[2023A](#), [gifs](#)].
○ Boosted deep hierarchical RL sample-efficiency by two-orders-of-magnitude [[2022D](#), [videos](#)].
○ Enhanced adversarial robustness of NN's with guarantees [[2022A](#)].
○ Developed out-of-distribution detectors with guarantees [[2022B](#)] that run in real-time [[2021B](#)].
- May - Aug 2023 **Amazon Web Services (AWS) AI Labs, Applied Scientist Intern,** *Santa Clara, CA.*
Hosts: [Dr. Abishek Sankararaman](#), [Dr. Vikram Nathan](#), [Dr. Murali Narayanaswamy](#)
○ Improved generalization in offline RL by incorporating transformer model based forecasts in conservative Q learning; applied to cloud resource allocation problems.

- May - Aug 2022 **Amazon Web Services (AWS) AI Labs**, *Applied Scientist Intern*, Santa Clara, CA.
 Hosts: [Dr. Abishek Sankararaman](#), [Dr. Murali Narayanaswamy](#)
 Accelerated datacenter resource allocation by combining model-free RL with mixed integer linear programs [[Preprint 2022C](#)].
- May - Aug 2021 **Argo AI (Ford & VW's Self-Driving Partner)**, *Systems Research Intern*, Dearborn, MI.
 Product Security and Sensor Functional Safety Team
 Built threat models for object detection and segmentation models on autonomous vehicles.

Selected Publications and Preprints

- [2025C] [SIMA 2: A Generalist Embodied Agent for Virtual Worlds](#)
 SIMA 2 team including **Kaustubh Sridhar**
 ➤ Technical Report, arXiv:2512.04797.
- [2025B] [SIMA 2: An Agent that Plays, Reasons, and Learns With You in Virtual 3D Worlds](#)
 SIMA 2 team including **Kaustubh Sridhar**
 ➤ Google Deepmind Blogpost.
- [2025A] [RICL: Adding In-Context Adaptability to Pre-Trained Vision-Language-Action Models](#)
Kaustubh Sridhar, Souradeep Dutta, Dinesh Jayaraman, Insup Lee
 ➤ Conference on Robot Learning (CoRL) 2025 (Acceptance rate: 35%).
- [2024A] [REGENT: A Retrieval-Augmented Generalist Agent That Can Act In-Context In New Environments](#)
Kaustubh Sridhar, Souradeep Dutta, Dinesh Jayaraman, Insup Lee
 ➤ International Conference on Learning Representations (ICLR) 2025,
 ➤ **Oral presentation** at ICLR 2025, **top 1.8% of 11672 submissions**,
 ➤ **NeurIPS 2024** workshops on Adaptive Foundation Models and Open World Agents.
- [2023B] [Memory-Consistent Neural Networks for Imitation Learning](#)
Kaustubh Sridhar, Souradeep Dutta, Dinesh Jayaraman, James Weimer, Insup Lee
 ➤ International Conference on Learning Representations (ICLR) 2024 (Acceptance rate: 31%).
- [2023A] [Guaranteed Conformance of Neurosymbolic \(World\) Models to Natural Constraints](#)
Kaustubh Sridhar, Souradeep Dutta, James Weimer, Insup Lee
 ➤ ICLR 2023 workshop on Neurosymbolic Generative Models,
 ➤ Conference on Learning For Dynamics and Control (L4DC) 2023.
- [2022D] [Exploring with Sticky Mittens: Reinforcement Learning with Expert Interventions via Option Templates](#)
 S. Dutta*, **K. Sridhar***, O. Bastani, E. Dobriban, J. Weimer, I. Lee, J. Parish-Morris
 ➤ Conference on Robot Learning (CoRL) 2022 (Acceptance rate: 39%).
- [Preprint 2022C] [Predict-and-Critic: Accelerated End-to-End Predictive Control for Cloud Computing through Reinforcement Learning](#)
Kaustubh Sridhar, Vikramank Singh[†], Murali Narayanaswamy[†], Abishek Sankararaman[†]
 ➤ Under review ([†]**Amazon AWS AI Labs**).
- [2022B] [CODiT: Conformal Out-of-distribution Detection in Time-series Data](#)
 Ramneet Kaur, **Kaustubh Sridhar**, Sangdon Park, Susmit Jha[†], Anirban Roy[†], Oleg Sokolsky, Insup Lee ([†]SRI International)
 ➤ ICML 2022 workshop on Principles of Distribution Shift,
 ➤ International Conference on Cyber-Physical Systems (ICCPS) 2023 (Acceptance: 25.6%),
 ➤ **Best paper award nomination** at ICCPS 2023.
- [2022A] [Improving Neural Network Robustness via Persistency of Excitation](#)
Kaustubh Sridhar, Oleg Sokolsky, Insup Lee, James Weimer
 ➤ American Control Conference (ACC) 2022.
- [2021B] [Real-Time Detectors for Digital and Physical Adversarial Inputs to Perception Systems](#)
 Yiannis Kantaros, Taylor Carpenter, **Kaustubh Sridhar**, Yahan Yang, Insup Lee, James Weimer
 ➤ International Conference on Cyber-Physical Systems (ICCPS) 2021 (Acceptance rate: 26%).

- [2021A] [Real-Time Data-Predictive Attack-Recovery for Complex Cyber-Physical Systems](#)
Lin Zhang, **Kaustubh Sridhar**, Mengyu Liu, Pengyuan Lu, F. Kong, Oleg Sokolsky, Insup Lee
➡ IEEE Real-Time and Embedded Technology and Applications Symposium (**RTAS**).
- [2019] [Finite-time, Event-triggered Tracking Control of Quadrotors](#)
Kaustubh Sridhar, Srikant Sukumar
➡ Conference on Guidance, Navigation and Control (**EuroGNC**) **2019**.

Awards

- 2026 [Charles Hallac and Sarah Keil Wolf Award](#) for **Best ESE Dissertation**
- 2025 **Oral Presentation (top 1.8% of 11672 submissions)** for REGENT [2024B] at ICLR 2025
- 2023 **Best Paper Award Nomination** for CODiT [2022B] at ICCPS 2023
- 2022 **Top Reviewer (top 10%)**, NeurIPS 2022
- 2022 **Outstanding Reviewer (top 10%)**, ICML 2022
- 2023 **NSF Travel Grant**, International Conference on Cyber-Physical Systems (ICCPS) 2023
- 2022 **Student Travel Grant**, American Control Conference 2022
- 2019 **The Dean's Fellowship**, University of Pennsylvania
- 2019 **The Howard Bradwell Fellowship**, University of Pennsylvania
- 2018 **SN Bose Scholarship**, Govt. of India and the Indo-U.S. Science and Technology Forum
- 2015 **KVPY Fellowship**, Govt. of India

Selected Talks

- Apr 2025 **Oral Talk at ICLR 2025** on [REGENT](#), our generalist embodied agent [[recording+Q&A](#)].
- 2024-2025 **Training Adaptive and Sample-Efficient Generalist Embodied Agents.**
- [Apr 2025] NTU Singapore
 - [Nov 2024] Google Deepmind
 - [Nov 2024] Apple MLR
- 2023 **Learning Better Policies and Dynamics Models with Memory-Consistent and Memory-Constrained Neural Networks.**
- University of Pennsylvania (GRASP Lab) [[video](#)]
 - University of Pennsylvania (Perception Action Learning Group)
- 2023 **Guaranteed Conformance of Neurosymbolic Generative (Dynamics) Models to Physics and Medical Constraints**
- Johns Hopkins University (CISS Session on Learning for Optimization and Control)
 - Amazon Science (Deep Earth Reading Group)
 - University of Pennsylvania (Formal Methods and Machine Learning Reading Group)

Press Coverage

- 2026 SIMA 2 Media Coverage: [The Verge](#), [MIT Tech Review](#), [TechCrunch](#), [The RunDown AI](#), [The Decoder](#), [WinBuzzer](#), [SiliconANGLE](#), [Forward Future](#), [The Tech Buzz](#), [Decrypt](#)
- 2023 [Making Better Decisions with AI](#), Penn Today (USA).

Service and Mentorship

- 2022 - Present **Reviewer**
[ICLR](#) 2025, 2024, [ICML](#) 2025, 2024, 2023, 2022, [NeurIPS](#) 2024, 2023, 2022, [L4DC](#) 2023, [ICCPS](#) 2025, 2022, [RSS](#) 2025
- 2020 - 2021 **Organizer**, *Reading Group in Robust Deep Learning*, University of Pennsylvania
- 2018 - 2019 **Team Lead**, *Department Academic Mentorship Program*, IIT Bombay
Led a team of 22 senior mentors to counsel 89 sophomores, 29 under-performing students.

Earlier Work Experience

- May - Jul 2018 **Duke University**, *Undergraduate Summer Research Fellow*, Durham, NC.
Advised by [Prof Miroslav Pajic](#), Cyber-Physical Systems Lab
◦ Developed a self-driving platform for intrusion detection testing [[videos](#)].
- Jan - Dec 2018 **Indian Institute of Technology Bombay**, *Undergraduate Research Assistant*, India.
Advised by [Prof Srikant Sukumar](#),
◦ Bachelor's thesis on real-time quadrotor control [[2019](#)].

Technical skills

- Languages Python, C, C++
Machine Learning Pytorch, JAX, OpenAI Gym, Tensorflow, CUDA, Sklearn, Pandas
Robotics Mujoco, Bullet, CARLA, ROS, Gazebo

Key Coursework

- Graduate Deep Learning, Reinforcement Learning, Convex Optimization, Probability, Computer Aided Verification
Undergraduate Data Structures and Algorithms, Linear and Nonlinear Control Theory, Adaptive and Optimal Control

Other Projects

- Apr - May 2023 "[Fixing Reward Hacking with Large Language Models](#)."
◦ Created a framework for an RL agent in Deepmind AI Safety environments to leverage GPT4 to detect reward hacking, fix its own reward function, and adapt quickly to the new reward.

Teaching

- Spring 22, Fall 24 **Teaching Assistant**, *CIS 541: Embedded Software for Life-Critical Systems*, UPenn
Spring 2021 **Teaching Assistant**, *CIT 595: Computer Systems Programming*, University of Pennsylvania